

APPLIED RESEARCH

Analyzing Information Leakage on Video Object Detection Datasets by Splitting Images Into Clusters With High Spatiotemporal Correlation

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ABSTRACT Random splitting strategy is a common approach for training, testing, and validating object detection algorithms based on deep learning. Is common for datasets to have images extracted from video sources, in which there are frames with high spatial correlation, i.e., frames with rotated positions or different view angles of the same object. These highly correlated frames may lead to information leakage in training, if these frames are not well-distributed. In this work, it is shown that datasets created with highly spatial correlation frames from the same video have information leakage if using the random splitting strategy to distribute the image into the sub-datasets. It proposed a clustering dataset split algorithm in which images are distributed randomly in the sub-datasets in a pack or clusters instead of a single image at the time. The clusters are created by extracting the image features from a video of the dataset using an image-text pre-trained model, CLIP, and reducing the feature vector dimensionality with t-Distributed Stochastic Neighbor embedding (t-SNE). In this reduced dimensional representation, images are separated into clusters using a clustering algorithms like DBSCAN, OPTICS, and Agglomerative Clustering. These clusters are distributed into the train, test, and validation datasets randomly to avoiding information leakage by highly spatial correlation frames. YOLOv8 is used as the object detector algorithm to test the dataset splitting.

INDEX TERMS Data preprocessing, clustering, information leakage, supervised training, video annotation.

I. INTRODUCTION

Object detection is been in a constant flow of improvements, from classification techniques to image segmentation with multiple objects [1], [2], [3]. It is a common practice to use videos as a source of data for the training phase in an object detection algorithms [4], [5], [6]. Video has a great amount of information about the object to be detected, possibly in the same scenario, and possibly in many angles and views. Thus, is a great source to improve the reliability and accuracy of object detection algorithms.

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Most datasets for object detection are made from images extracted from a video source. A video scene may contain many frames with high spatiotemporal correlation, not only with adjacent frames but in frames that have repeated movements along the video. These frames may not be identical but may have a high similarity, and this may indicate that there is significant spatial redundancy present in most frames. As an example, four images from the same scene are presented in the Fig 1. In the four images, all elements, background, and position of the camera are the same, i.e., the spatiotemporal correlation is high. This observation indicates that current benchmark datasets may have obvious biases in terms of moment location distribution, and it is hard to judge whether existing methods are merely fitting the biases or truly

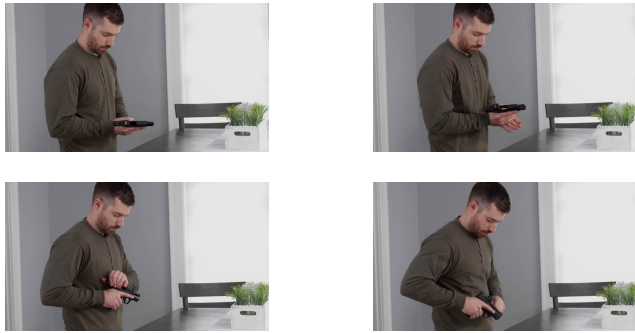


FIGURE 1. Example of 4 images extracted from the same scene of a video.

learning the semantic alignment relationship between the two modalities [7].

Generally, to train a deep learning object detection model, the data is split into sub-datasets, which are used to train, validate and test the model. A common division is to split the dataset into three different parts, training, validating and testing. Training sub-dataset is used to fit the model parameters, the validation takes a role of tuning the model hyper-parameter and to give an estimate of the skill level of the model, withal testing dataset is used for estimating the final tuned model skill level [8]. More importantly, it is widely accepted that no overlap should exist between the samples used for model fitting and hyperparameter tuning, and those belonging to the testing set [9].

Most deep learning object detection algorithms use random sampling to train and test, but due to the temporal redundancy that frames from a video may have, some information about the test may be leaked. This affects the generalization ability of the model, making the model only learn the distribution of one domain [10].

In this work, is proposed a general methodology to find information leakage between training, validation and testing datasets generated by the random split technique. Is proposed a new split technique to demonstrate the existence of information leakage called cluster split. The technique split the data into three sub-datasets, making that every frame, from the same video, with high similarity will be allocated in the same sub-dataset. The proposed methodology may also apply to different types of data in other supervised learning circumstances.

This work is organized as follows, section II presents the related works, where similar dataset split algorithm are proposed. In Section III presents the proposed algorithm in detail. Section IV shows every component used in the experiment, in IV-A details the dataset, IV-B the feature image extractor, IV-C the algorithm used in dimensionality reduction, IV-D presents the clustering algorithms, used as comparison, to separate the clusters. IV-E presents the object detection model, IV-F the metrics used to analyse the results, and IV-G the experiment setup. A conclusion and future works is drawn in VI.

II. RELATED WORK

Hänsch show in [11] that sampling strategies applied to partition the available reference data into train and test sets have a large influence on the quality and reliability of the estimated generalization error. It illustrates and discusses problems of common choices for sampling schemes, i.e. the violation of the independence assumption and the illusion of the availability of global knowledge in the training data. It propose a novel dataset split, denoted as Cluster Sampling, which consist that for each class the spatial coordinates of all samples are clustered into two clusters. Training samples of a class are randomly drawn from one of the clusters, the other cluster is used as test data.

Yagis in [12] quantitatively evaluates the effect of data leakage cause by 3D magnetic resonance imaging (MRI) data splitting based on a 2S slice-level using three 2D CNN models to classify patients with Alzheimer's disease (AD) and Parkinson's disease (PD). It is shown that exist an inflation on the accuracy using the slice-level cross-validation on the test set by 30% on Open Access Series of Imaging Studies (OASIS), 29% on Alzheimer's Disease Neuroimaging Initiative (ADNI), 48% on Parkinson's Progression Markers Initiative (PPMI) and 55% on a local de-novo PD Versilia dataset.

In [8] Kahloo proposes an algorithmic method to prepare the sub-dataset splits for machine learning models, for evenly representing the dataset in training and validation sub-datasets bases on the dataset distribution. The method consist in representing the dataset in a lower dimension and to cluster the dataset into three separate ranges (median range, quartile range, and extreme range) for each cluster and gathering each range solely is done. Two types of clustering are examined, (HDBSCAN) [12] and Gaussian Mixture Models (GMM) [13]. Building a subset by picking data from these groups separately then combining them will reduce the randomness, and will ensure that the resulted subset will contain data distribution closer to the original data distribution.

In [13] Roshan proposes a method called SPlit for data splitting based on the method of support points (SP), which is adapted to subsampling from a dataset using a sequential nearest neighbor algorithm. They are then used as the testing set and the remaining as the training set. We have also extended the method of SP to deal with categorical variables. Thus, SPlit can be applied to both regression and classification problems.

In [9], Tampu demonstrate the effect of improper dataset splitting on model evaluation for three classification tasks using three optical coherence tomography (OCT) open-access datasets extensively used, Kermany's and Srinivasan's ophthalmology datasets, and AIIMS breast tissue dataset. Results show that the classification performance is inflated by 0.07 up to 0.43 in terms of Matthews Correlation Coefficient (accuracy: 5% to 30%) for models tested on datasets with improper splitting, highlighting the considerable effect of dataset handling on model evaluation.

A novel evaluation protocol is proposed in [7] by Lan et al., and a dataset re-splitting based on the data-set momentum and distributions characteristics, each dataset is re-split into four sets, training, validation, test-iid (related to independent and identical distribution) and test-ood (related to out of distributions). The proposed, claims a methodology to eliminate conflicting videos on dataset, avoid performance inflation.

Feng [10] made a survey about information leakage in hyperspectral remote sensing image classification tasks. The work gathered some strategies to mitigate the effect of information leakage for spatial disjoint sampling, like a Cluster Sampling Strategy [11] that for each class, it is clustered into two clusters according to its spatial coordinates. The training samples are randomly selected from a cluster, and the remaining classes are used as the test set. Another strategy cited is the Density-Based Clustering Algorithm Sampling Strategy: Lange et al. [14] that detect subgroups in a set by recursively evaluating the density threshold of neighbor points around the sample with parameter ϵ as the search radius. Therefore, independent regions can be determined by clustering the coordinates of pixels of a particular class.

To the best of our knowledge, most of the related work are focused on classification problem, to evaluate the effect of dataset splitting in areas that the samples images are very similar. The novelty of this work is to evaluate the effect of dataset splitting in object detection problem.

III. PROPOSED METHOD

This paper presents an algorithm to split an image dataset with images extracted from video sources, into training, validation, and test sub-datasets, in a manner that similar images derived from the same video will be allocated in the same sub-dataset. The main objective is to show that random splitting techniques may have information leakage caused by similar images, inflating the performance metrics. The full dataset may contain images extracted from any sources, but sources like a video, which contains images with high similarities, will be separated differently from other sources. Each video is visually analyzed to create the ground truth clusterization labels.

The ideal case scenario is to identify every frame with high spatiotemporal correlation from the same video, clusterized it, and move all images from the same cluster to the same sub-dataset. This will ensure that images with a strong spatial-temporal correlation will be set on the same sub-dataset, mitigating the effect of information leakage. To achieve this goal, the algorithm can be split in the following steps:

- First, a dimensionality reduction step is performed to facilitate the similarity process of the dataset. Features are extracted from every image using the image-text pretrained model CLIP, which reduces the images to 512 embedded vector. To represent the images in a

two-dimensional space, it is applied a dimensional reduction algorithm (t-SNE).

- Apply a clusterization algorithm on the assigned data points. This clusterization needs to split images with high spatiotemporal correlation in the same cluster.
- The dataset is randomly split into sub-dataset by moving all images in the same cluster to the same sub-dataset. Instead of moving one image at a time, the whole cluster is set to a determined sub-dataset.

The algorithm can be summarized by adding a step before to the random split algorithm, which consists of treat the input of the random split as a cluster instead of a single image.

A cluster is defined as a set of images from the same video with high spatiotemporal correlation. Moving a cluster with size N to a specific sub-dataset is like moving N images. The main idea is to cluster the images, extracted from a video, in a manner to achieve the maximum number of cluster M , and maintain images with high similarities in the same cluster. The maximization of M increases the randomness in the division of the sub-datasets. Lower values of M will give few possibilities of randomizing the sub-datasets, which can give some kind of biases.

The overall structure of the algorithm is presented in Fig. 2 for a single video, and the process is repeated for every video with high spatial-temporal correlation. For each video, a dimensionality reduction is made, and the clusterization is also realized in this reduced dimension. After that, every scene can be randomly selected into sub-datasets.

IV. EXPERIMENTS

The proposed algorithm is experimented in a single object detection dataset, a feature extraction model, a dimensionality reduction algorithm, and applying different clustering methods to evaluate the performance against the traditional random split technique [15]. The dataset, feature extraction, clustering methods, trained models, and the designed experiments are explained more in this section.

A. DATASET

The dataset created for this task is an object detection dataset with a single class, handguns. This dataset is composed of only images taken from videos. The dataset is composed of 1490 images selected from YouTube videos with dimensions 1920×1080 . The dataset is separated into three sub-datasets, train, validation, and test. The sub-datasets have sizes of 80%, 10%, and 10% of the number of images.

The annotations of the dataset were created with CVAT(Computer Vision Annotation Tool [16]) tool. CVAT is a free, open-source, web-based image and video annotation tool that is used for labeling data for computer vision algorithms.

The videos chosen to extract images are separated as scenes. A scene is defined as a set of images with high spatiotemporal correlation, i.e., that are recorded in the same

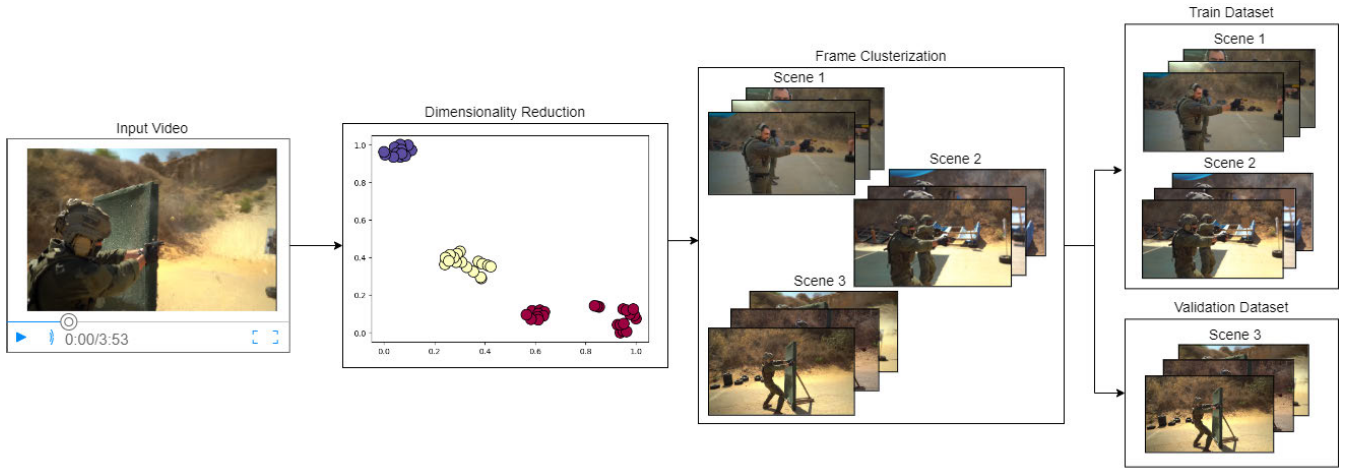


FIGURE 2. Overall diagram of splitting images of a video into sub-datasets.

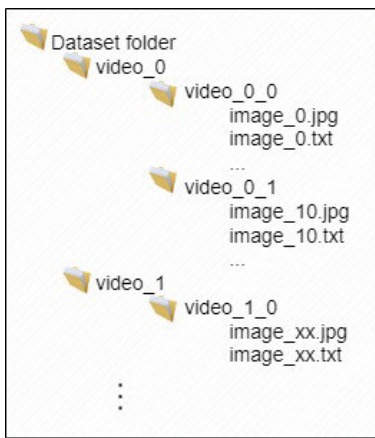


FIGURE 3. The folder structure of the dataset.

background, containing the same objects but in different positions or rotations.

A ground truth (GT) clusterization is created manually for every video. This GT clusterization is used for comparison with the scene clusterization created by the cluster algorithms. The structure of the dataset is presented in Fig 3. All images extracted from the videos used in the dataset are placed in the same folder, and inside each video folder is placed every scene separated by the chosen cluster algorithm into sub-folders. Inside each sub-folder, it is placed all the images from the scene with the labels, both with the same name.

B. FEATURE EXTRACTION ON IMAGES

Digital images can be represented as three matrices with $N \times M$ dimensions, they can be classified as high-dimensional data. Each pixel is an image element at a unique location with an associated attribute vector. There is often lots of redundant information in it, including related or duplicated factors.

One manner to reduce the dimensionality representation of an image is using feature extraction, which is the process of transforming raw data into numerical features that can be processed while preserving the information in the original data set. A vector of features (embedded vector) can be extracted using an AI pre-trained model, for example, CLIP(Contrastive Language-Image Pre-Training) [17]. CLIP is a neural network trained on a variety of (image, text) pairs. We use CLIP's official augmentations and extract vision features from its image encoder. The extracted features, for each image, are reduced to 512 vector length, defined as an embedded vector. Because the size of the embedded vector dimension is still very large, it is needed to reduce it to a two-dimensional vector to visually analyze it.

C. DIMENSIONALITY REDUCTION

The embedded vector created by Clip reduces the dimensionality of the data, but to simplify the distance calculation and visually analyze the data, we reduce even more the data representation even to fit in a two-dimensional space. This reduction is realized using the t-Distributed Stochastic Neighbor Embedding (t-SNE) [18].

t-SNE is a technique that visualizes high-dimensional data by giving each data point a location in a two or three-dimensional map, it provides an effective technique for visualizing data and dimensionality reduction, which is capable of preserving local structures in high-dimensional data while retaining some global structures. t-SNE utilizes a joint probability distribution to model the similarity in high-dimensional space

$$p_{i,j} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\delta^2)}{\sum_{k \neq l} \exp(-\|x_l - x_k\|_2^2 / \delta^2)}, \quad (1)$$

where δ is the variance parameter of Gaussian. Moreover, to eliminate the ‘‘Crowding Problem’’, t-SNE employs Student t-distribution with one degree of freedom to model

the similarity between data y_i and y_j in low-dimensional space [19],

$$q_{i,j} = \frac{(1 + \|y_i - y_j\|_2^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|_2^2)^{-1}}. \quad (2)$$

t-SNE finds the optimal low-dimensional representations for matching p_{ij} and q_{ij} to the greatest extent. This is achieved by minimizing the following Kullback-Leibler divergence measuring the difference between two probability distributions,

$$C(Y) = KL(P \parallel Q) = \sum_i \sum_j p_{i,j} \log \frac{p_{i,j}}{q_{i,j}}. \quad (3)$$

t-SNE calculates the optimal low-dimensional representation Y by minimizing $C(Y)$ over all data points with a gradient descent method [19]. The algorithm starts by applying SNE (Stochastic Neighbor Embedding) to the data points, which converts the high-dimensional Euclidean distances between data points into conditional probabilities that represent similarities [18].

D. CLUSTERING ALGORITHMS

Clusterization is one of the main parts of the proposed algorithm. Clustering videos is a hard task because there is no information about the number of clusters (scenes) or about the shape of the clusters. These clusterization algorithms are not expected to be ideal, but the purpose of using them is to observe the impact of information leakage in sub-datasets with bigger amounts of similar images, and observe the changes in performance metrics. To be able to glimpse the effect of information leakage made by image similarity, a ground truth clusterization is used as a comparison to analyze the performance of the clusterization algorithms.

1) DBSCAN

The DBSCAN algorithm views clusters as areas of high density separated by areas of low density. Due to this rather generic view, DBSCAN finds clusters with various formats easily [20]. The central component for DBSCAN is the concept of core samples, which are samples that are in areas of high density. A cluster is therefore a set of core samples, each close to each other, measured by some distance measure and a set of non-core samples that are close to a core sample, but are not themselves core samples.

The algorithm works as follows. A central sample is determined and then other samples are traversed, for all unmarked samples (not part of a cluster), distance metrics are used to define whether this sample is part of the same cluster as the central sample, these distance metrics take into account both the central sample and the other remaining samples. A temporary result is saved if the distance metrics match the sample being densely close to the central sample. The process of finding central samples is based on the minimum amount of samples that must be densely connected to this one [20]. Central samples and their groups are then merged depending

on the amount of common samples between them. Finally, the result delivers dense clusters separated by empty spaces between them.

2) OPTICS

The OPTICS (Ordering Points To Identify the Clustering Structure) algorithm, which is derivative from the DBSCAN, was introduced by [21]. The algorithm does not produce a clustering of a dataset explicitly, instead, it outputs an augmented ordering of the data representing its density-based clustering structure. Because of that, the output contains information that is equivalent to density-based clustering corresponding to a wide range of parameter settings and is, therefore, more versatile for clustering analyses that need to be automated [21].

3) AGGLOMERATIVE CLUSTERING

The Agglomerative Clustering works in recursion by merging pair of clusters of sample data based on a linkage distance metric [22]. The Scikit-learn implementation of the algorithm defines that it performs a hierarchical clustering using the bottom up method: each sample starts in its own cluster and the clusters are successively merged. The possible linking criteria, which determine the metric used for the merge strategy, are [23]:

- Ward minimizes the sum of squared differences across all clusters. Its objective is to minimize variance, thus being similar to the k-means objective function, but revised as an agglomerative hierarchical approach.
- Maximum or complete linkage minimizes the maximum distance between samples of pairs of clusters.
- Average linkage minimizes the average of the distances between all samples of cluster pairs.
- Single linkage minimizes the distance between the closest samples of cluster pairs.

E. OBJECT DETECTOR

The model selected to use as object detector is the YOLOv8 [24]. YOLOv8 is a series of improvements and extensions made by Ultralytics to the YOLOv5 architecture, presented in [25]. YOLO-V8 achieves state-of-the-art performance through model structure optimization, optimization of the anchor box or anchor free scheme, and various data augmentation techniques. It is trained to detect a single class, handguns, using the pre-trained weights (S-model YOLO-V8).

F. EVALUATION METRICS

The evaluation of the cluster split algorithm can be separated in three parts. First part is to analyze the representation of the dataset in a lower dimension, i.e., and embedding vector. In this form, the similarity between images is easily computed. One form to measure the similarity between embeddings is the cosine similarity metric, which is a measure of similarity between two vectors of an inner

product space that measures the cosine of the angle between them [26]. The cosine similarity is commonly used to measure the similarity between embedded vectors [27], [28], [29]. Without loss of generality, defining two image embedded vectors as query, $\mathbf{I}_q$, and reference as $\mathbf{I}_r$, the cosine similarity (CS) is defined as

$$CS(\mathbf{I}_r, \mathbf{I}_q) = \frac{\mathbf{I}_r^\top \mathbf{I}_q}{\|\mathbf{I}_r\| \|\mathbf{I}_q\|}. \quad (4)$$

Is it possible to measure the similarity between sub-datasets and represent it as a similarity matrix $\mathbf{s} \in \mathbb{R}^{n_q} \times \mathbb{R}^{n_r}$, where n_q is the number of images in reference sub-dataset, and n_r is the number of images on the query sub-dataset. The i -th image from the reference dataset is denoted as $\mathbf{I}_q^i$ and the j -th image from the query dataset is denoted by $\mathbf{I}_r^j$. The similarity matrix is defined as

$$s_{ij} = CS(\mathbf{I}_r^i, \mathbf{I}_q^j). \quad (5)$$

The second part is related to cluster generation. To measure the clustering performance, we use two well-known clustering metrics, the Adjusted Rand Index (*ARI*), and Adjusted Mutual Information (*AMI*). *ARI* takes into account the number of instances that exist in the same cluster and different clusters. The expected value of such validation measure is not zero when comparing partitions [30]. *ARI* is defined as

$$ARI = \frac{n_{11} + n_{00}}{n_{00} + n_{01} + n_{10} + n_{11}} = \frac{n_{11} + n_{00}}{\binom{n}{2}}, \quad (6)$$

where n_{11} is the number of pairs of instances that are in the same cluster in both, n_{11} the number of pairs of instances that are in different clusters, n_{10} the number of pairs of instances that are in the same cluster in A , but in different clusters in B , and n_{01} , the number of pairs of instances that are in different clusters in A , but in the same cluster in B . The value of *ARI* lies between 0 and 1, and the higher value indicates that all data instances are clustered correctly and the cluster contains only pure instances.

AMI is one of the common external clustering validation metrics that estimate the quality of the clustering with respect to a given class labeling of the data, it is derived from the Mutual Information Score (*MI*), the need to this adjusted version, comes from the fact that the original metric increases with the number of samples of clusters, regardless of the actual information shared. More formally, *AMI* can effectively measure the amount of statistical information shared by random variables representing the cluster assignments and the pre-defined label assignments of the instances [30], [31], [32]. Thus, *MI* and *AMI* is calculated as follows:

$$MI(U, V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} \frac{|U_i \cap V_j|}{N} \log \left(\frac{N \times |U_i \cap V_j|}{|U_i| |V_j|} \right) \quad (7)$$

In equation 7 above, $|U_i|$ and $|V_j|$ represents the number of samples in each cluster respectively, the N symbol represents

the total number of samples of all clusters.

$$AMI(U, V) = \frac{[MI(U, V) - E(MI(U, V))]}{[avg(H(U), H(V)) - E(MI(U, V))]} \quad (8)$$

In the equation 8 the symbols *avg*, *E* and *H*, represents the average, the expectation and the Shannon's entropy.

The last part, is the set of metrics needed to evaluate the object detector. The metric mean Average Precision (*mAP*) is extensively used in conventional object detection, which provides a performance evaluation in terms of regression and classification accuracy [4]. Before examining the variations of the *AP*, we should review some concepts that are shared among them. The most basic are the ones defined below:

- True positive (TP): A correct detection of a ground-truth bounding box;
- False positive (FP): An incorrect detection of a non-existent object or a misplaced detection of an existing object;
- False negative (FN): An undetected ground-truth bounding box.

It is important to note that, in the object detection context, a true negative (TN) result does not apply, as there are infinite number of bounding boxes that should not be detected within any given image. To be considered a correct detection, the overlap area between the predicted bounding box B_p and the ground truth box B_{gt} have to be greater than a threshold, this can be calculated using the Intersection Over Union (IOU), a measure based on the Jaccard Index, a coefficient of similarity for two sets of data.

$$a_o = \frac{\text{area}(B_p \cap B_{gt})}{\text{area}(B_p \cup B_{gt})}, \quad (9)$$

where $B_p \cap B_{gt}$ denotes the intersection of the predicted and ground truth bounding boxes and $B_p \cup B_{gt}$ their union. the *mAP* can be defined depending on how the threshold is chosen:

- *mAP@.5*: mean Average Precision at IOU threshold of 0.5. Since the IOU threshold of 0.5 is relatively low, this evaluation method focuses more on initially finding an object [33].
- *mAP@.5 : .95*: mean Average Precision at IOU threshold of 0.5 to 0.95 with an interval of 0.05, also named COCO *mAP*. Since *mAP@.5 : .95* averages the precision under different IOU thresholds, compared to *mAP@.5*, *mAP@.5 : .95* focuses more on the ability of precisely locating an object.

Since, as stated above, the true negatives (TN) are not used in object detection frameworks, one refrains to use any metric that is based on the TN, such as the TPR, FPR and ROC curves [34]. Instead, the assessment of object detection methods is mostly based on the precision *P* and recall *R* concepts, respectively defined as

$$P = \frac{TP}{TP + FP} = \frac{TP}{\text{all detections}}, \quad (10)$$

$$R = \frac{TP}{TP + FN} = \frac{TP}{\text{all ground truths}}. \quad (11)$$

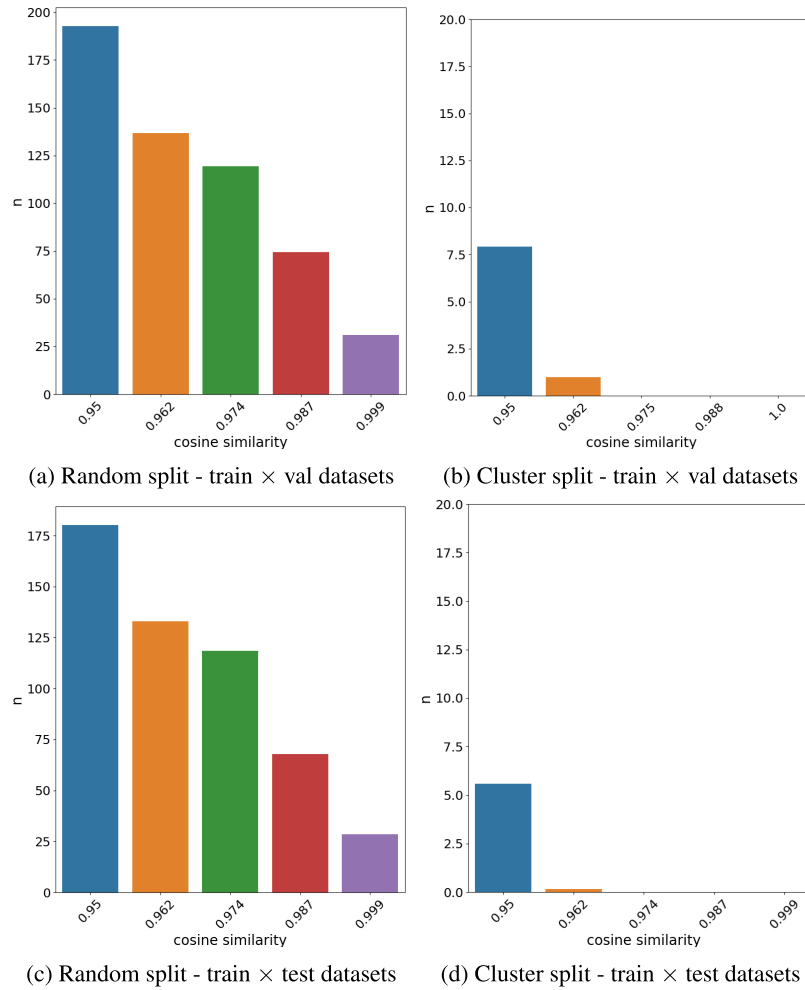


FIGURE 4. Cosine similarity histogram for image comparison between sub-dataset of the same split technique.

Precision is the ability of a model to identify only relevant objects. It is the percentage of correct positive predictions. Recall is the ability of a model to find all relevant cases (all ground-truth bounding boxes). It is the percentage of correct positive predictions among all given ground truths [34].

G. EXPERIMENT SETUP

For each splitting algorithm, 15 training cycles were performed and split into three sub-datasets, train, validation, and test sub-datasets. To establish a baseline, ground truth clusterization is used, which is supposed to represent the best way of splitting the data. It also generates a random split on the dataset using the traditional random algorithm and repeats 15 times. All training is made separately, i.e., there are no relations between the training rounds for the same algorithm. Each dataset split is used as an input for the object detector using yolov8, which has the same parameters configuration, 100 epochs, and image size of 640. All simulations are realized with Intel(R) Xeon(R) Gold 6252 CPU-2.10GHz, 10 Gb of RAM memory, and 10 Gb GPU memory.

V. RESULTS AND DISCUSSION

In this work it is studied only the effect of data leakage by very similar images. However, there are other sources of data leakage stemming from hyper-parameter optimization, data normalization, and data augmentation [9].

A. IMAGE SIMILARITY ANALYSIS

This section presents the relation of image similarity contained between sub-datasets. It is compared the number of similar images in the sub-datasets for the random split and the cluster split, specifically, between the train and validation, and train and test sub-datasets, i.e., all images contained in the train sub-dataset are compared with all images contained in the validation sub-dataset. The similarity is measured by the cosine similarity, and a similarity matrix is made showing the relation between two sub-dataset. Several clusterization techniques are used for the cluster split in this work, for the sake of simplicity, this investigation will use only the GT clusterization and the random technique.

Fig 4 shows the cosine similarity matrix histogram for values of cosine similarity between $\{0.95, 0.99\}$,

TABLE 1. Best parameters for all clustering algorithms.

Algorithm	AMI	ARS	Parameters
DBSCAN	0.633	0.620	$eps = 0.045$ $min_samples = 2$
OPTICS	0.295	0.049	$min_samples = 2$ $max_eps = 0.03$ $metric = chebyshev$ $leaf_size = 2$
Agglomerative Clustering	0.390	0.083	$n_clusters = None$ $distance_threshold = 0.08$

TABLE 2. Mean values of bests results on validation data for clustering algorithms.

Algorithms	Precision	Recall	$mAP@.5$	$mAP@.5 : .95$
Ground Truth	0.873	0.750	0.834	0.545
Random	0.958	0.922	0.965	0.765
DBSCAN	0.898	0.785	0.870	0.617
Optics	0.951	0.901	0.952	0.736
Agglomerative	0.928	0.859	0.908	0.655

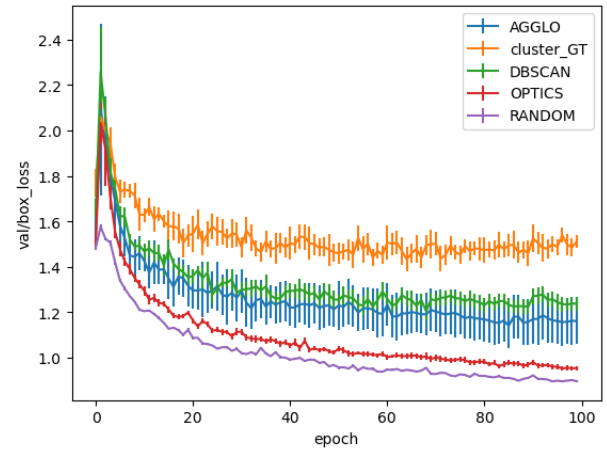
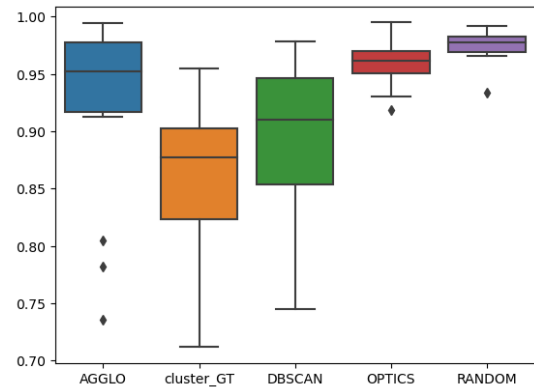
and these histograms are the mean of the cosine similarity for 15 different splits for both split techniques. Fig 4a shows the histogram for cosine similarity matrix between the train and validation sub-datasets for the random split technique and Fig 4b the same histogram for train and validation sub-dataset for cluster split technique. It is perceptible that the random split technique has more pairs of similar images (544) than the cluster technique (8.9). The same pattern occurs for the comparison between the train versus test datasets, the random split technique has more pairs of similar images (528) than the cluster technique (5.75).

B. CLUSTERIZATION PARAMETERS AND METRICS

In this section is presented the best clusterization parameters found for the analyzed algorithms for AMI and ARS metrics. This results have a lead on how much the information leak will be expected on the performance of the object detector. In Table 1 it is shown the best parameters found for each cluster algorithm, and the results presented are the mean for all analyzed videos. Also, the best result are found by setting the t-SNE perplexity parameters to 20. For this test environment, DBSCAN algorithm has the better results, followed by the Agglomerative Clustering, and last the OPTICS.

C. OBJECT DETECTION RESULTS

The performance of the Yolov8 object detector together with the dataset split strategies is summarized in Table 2, for the metrics Precision, Recall, $mAP@.5$, and $mAP@.5 : .95$ metrics. The results presented are the mean for the best result for the 15 train realized. Results on the random split show that there is an inflation on the metrics in comparison with other splitting strategies. In comparison with the GT clusterization, the random split has 14%, 22%, 15%, and 40% results for Precision, Recall, $mAP@.5$, and $mAP@.5 : .95$. It is possible to see that the metrics used for object detection


FIGURE 5. Bounding Box loss for model training.

FIGURE 6. Box plot for $mAP@.5$ metric.

($mAP@.5$, and $mAP@.5 : .95$) have a major impact on the information leakage, and a possible reason is that the spatial information for the objects are been leaked between sub-dataset.

Yolov8 algorithm shares some interesting metrics losses during training. Fig 5 shows the bounding box loss on the validation process. Bounding box loss calculates the error between the predicted and the ground truth boxes' geometry. It measures how well the model predicts the size and location of the bounding boxes. The lower losses are from the random split and the higher values are from the GT cluster. Reiterating that the only difference between the techniques is the dataset split, this difference on bounding box loss is related to the information leakage in the sub-dataset. Those sub-datasets with lower bounding box loss are those with a high amount of similar images. The loss for the clustering technique is not dropping, which means that the model is not learning more, and maybe is needed more data to the model get better results. In contrast with the random technique, the loss gets lower until the end of the train. For models with a higher amount of information leakage, i.e., with a higher amount of similar images, the loss has a smooth decrease until the end of the train.

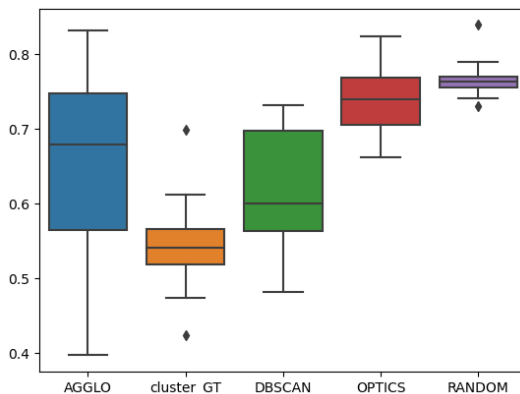


FIGURE 7. Box plot for $mAP@.5 : .95$ metric.

Fig 6 shows the boxplot of the best $mAP@.5$ results in each clustering technique. For all clustering algorithms, the variance is higher in comparison with the random split, and the same situation is seen for the metric $mAP@.5 : .95$ in Fig 7. This high variance, also seen in the loss, indicates that the model is not generalizing well with techniques with lower information leakage. The model may have not enough information about the object and need more epochs of training or more data.

VI. CONCLUSION AND FUTURE WORK

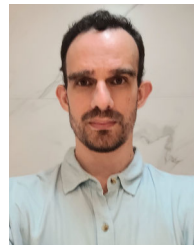
In this work, it is proposed a method for dataset split, based on clusterization of image with high spatio-temporal correlation extracted from videos, to diminish the effect of information leakage. The method extract every image feature from the dataset and cluster them using a cluster algorithm, and split the cluster into sub-datasets. It have been shown that the random splitting algorithm contains some information leakage in sub-dataset when images with high spatial correlation are not well separated. Also, it is presented results for cluster splitting with well known cluster algorithm, even though, in this algorithms, has some information leakage between sub-datasets, the results have a significant impact.

For future works, is necessary more study to metrify the impact of information leakage in object detection training. We propose to use a cluster algorithm that has better performance results, and be capable of generalizing the clusterization for every video. Expand this analysis to other type of data, i.e., implement the cluster algorithm to others dataset that have high correlation between samples. Also, generate the cluster from a 3d object, rotating the object and getting to get all information possible without any leakage. This may lead to object detection models with lower flickering on inference made on videos.

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